

Contact

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Top Skills

Big Data Analytics
Apache Spark
Spring Boot

Ramu Ramaiah

Software Architect at IBM
Greater Bengaluru Area

Summary

Software Architect with 20+ years architecting and scaling enterprise middleware, integration, and cloud platforms used across large global customer bases. Proven track record setting long-term technical strategy and architecture roadmaps, leading org-wide platform transformations (monolith-to-microservices, data-lake and streaming architectures), and driving adoption of emerging technology — from Big Data and Machine Learning to modern agentic AI frameworks. Deep, hands-on technical credibility across Java, Scala, and Python, paired with a track record of mentoring engineering teams, running architecture governance, and partnering directly with product leadership to translate business strategy into scalable technical direction.

Core competencies:

- Enterprise & Platform Architecture
- Large-scale distributed systems, microservices & cloud-native design, API/platform strategy, architecture governance
- Data & AI
- Big Data & streaming (Apache Spark, Kafka, Elasticsearch, Parquet), Machine Learning, applied/agentic AI (LangChain, LangGraph, PydanticAI)
- Engineering Leadership
- Technical roadmap ownership, cross-functional stakeholder management, mentoring, design & code review leadership
- Languages & Platforms
- Java, Scala, Python; performance & scalability engineering; high availability & fault-tolerant system design

Experience

IBM
Software Architect
July 2024 - Present (2 years 3 months)
Bengaluru

- Own the technical architecture and roadmap for the platform's data modernization initiative: architected an ETL pipeline (Apache Spark) migrating archived transactional data from an Entity-Attribute-Value model to a column-oriented model, unlocking self-service analytics for any off-the-shelf BI tool.
- Architected "Document Tracer," a graph-based document-lineage engine linking Purchase Orders, Invoices, and Shipment Notifications end-to-end — establishing a reusable architectural pattern for cross-document traceability across the B2B platform.
- Defined the architecture and led implementation of an AI-driven Partner Onboarding capability built on an in-house agentic framework layered on LangGraph, positioning the platform as an early adopter of applied AI within enterprise B2B integration.

Software AG

10 years 7 months

Architect

May 2019 - July 2024 (5 years 3 months)

Bangalore

- Architected a machine-learning-based recommender engine embedded in the webMethods Flow low-code environment, applying data-mining techniques (co-location, co-occurrence via Spark/Mahout) with an Elasticsearch-backed serving layer to lift developer productivity platform-wide.
- Led a foundational re-architecture of the platform's core in-memory data structures — including a custom "unrolled linked list" implementation — cutting memory consumption 30–40% across the entire product line, a systemic performance improvement with company-wide impact.
- Directed a build-vs-adopt architecture decision on JSON Schema validation, retrofitting the Everit parser onto Jackson; avoided a costly ground-up rebuild and delivered material engineering cost savings.
- Owned the high-level architecture of webMethods Data Hub, an in-house data lake paired with BI tooling (Dremio): designed a Kafka-based streaming pipeline transforming multi-GB daily transactional volume into Parquet columnar storage for analytical SQL workloads.
- Extended core platform capability by introducing WebSocket protocol support into webMethods Integration Server, built on Jetty's WebSocket implementation.
- Spearheaded a multi-year architectural transformation of the Integration Server from a monolith to a microservices architecture, introducing resiliency patterns (circuit breaker, bulkhead) and modernizing core modules around Google Guice-based dependency injection.

- Acted as the platform's technical mentor and architecture review lead — owning high-level design and code review, and partnering directly with product management to translate business goals into functional and non-functional requirements spanning availability, scalability, and fault tolerance.

Principal Engineer

January 2014 - July 2024 (10 years 7 months)

Bangalore

- Worked on cloud enabling the webMethods Integration Server, cloud with on-premise application integration.

Ariba

Senior Consultant Engineer

March 2011 - January 2014 (2 years 11 months)

- Involved in design and development of Collaboration/Proposal module and the ability for the temporary labors and contractors to enter time sheets for the hours they have worked on. The time sheets later will be converted to Invoices for payment.
- Involved in the Procurement Infrastructure area where the commonalities of Procurement and Supplier Management are handled. The Procurement Infrastructure area handles user management, supplier management, accounting management and approval process management. I am responsible for adding new enhancements to the module and maintain the existing module.

Software AG

Lead Engineer

May 2007 - February 2011 (3 years 10 months)

- Responsible for leading team efforts in requirement analysis, writing product functional specifications and developing new features in web services stack.
- Improve technical architecture, experiment with new software technologies and prototyping new modules based on new ideas and technology.

Motorola

Senior Software Engineer

July 2005 - April 2007 (1 year 10 months)

Motorola's telecom application stacks use a common High Availability solution that is developed completely in-house. The high availability platform includes

various modules that include platform abstraction layer, clustering, diagnostic subsystem, messaging etc.

- I was involved with clustering, diagnostic subsystem and messaging modules.

Honeywell

Senior Software Engineer

June 2001 - July 2005 (4 years 2 months)

- I was with Primus EPIC integrated Avionics Platform group. The Primus EPIC platform is an integrated Avionics system to provide all the avionics related functionalities. This includes flight management system, communication and navigation system. The platform provides core functionality needed for all the application stacks.
- I was also involved with the On-board maintenance system group for Airbus A380. The On-board maintenance system is designed to capture all the main events that happen during the air and display it on ground for regular maintenance services.

Education

Coimbatore Institute of Technology

· (1997 - 2001)

MAMHSS School